



# Text Preprocessing on 20 Newsgroups Dataset using NLTK

```
In [42]: import os
import pandas as pd
import nltk
import string

from nltk.corpus import stopwords
from nltk.tokenize import word_tokenize
from nltk.stem import WordNetLemmatizer
```

```
In [35]: nltk.download('punkt')
nltk.download('stopwords')
nltk.download('wordnet')
```

```
[nltk_data] Downloading package punkt to
[nltk_data] C:\Users\sabes\AppData\Roaming\nltk_data...
[nltk_data] Package punkt is already up-to-date!
[nltk_data] Downloading package stopwords to
[nltk_data] C:\Users\sabes\AppData\Roaming\nltk_data...
[nltk_data] Package stopwords is already up-to-date!
[nltk_data] Downloading package wordnet to
[nltk_data] C:\Users\sabes\AppData\Roaming\nltk_data...
[nltk_data] Package wordnet is already up-to-date!
```

Out[35]: True

```
In [36]: !kaggle --version
```

```
'kaggle' is not recognized as an internal or external command,
operable program or batch file.
```

```
In [37]: data_dir = "data"

texts = []

for file in os.listdir(data_dir):
    if file.endswith(".txt"):
        file_path = os.path.join(data_dir, file)
        try:
            with open(file_path, 'r', encoding='latin1') as f:
                texts.append(f.read())
        except:
            pass

print("Total documents loaded:", len(texts))
```

Total documents loaded: 20

```
In [38]: df = pd.DataFrame({'text': texts})
df = df.dropna().head(5000)
```

```
df.head()
```

Out[38]:

|   | text                                              |
|---|---------------------------------------------------|
| 0 | From: mathew <mathew@mantis.co.uk>\nSubject: A... |
| 1 | From: lipman@oasys.dt.navy.mil (Robert Lipman)... |
| 2 | Newsgroup: comp.os.ms-windows.misc\ndocument_i... |
| 3 | Newsgroup: comp.sys.ibm.pc.hardware\ndocument_... |
| 4 | Newsgroup: comp.sys.mac.hardware\ndocument_id:... |

In [44]:

```
def clean_text(text):  
    text = text.lower()  
    text = text.translate(str.maketrans('', '', string.punctuation))  
    return text  
  
df['clean_text'] = df['text'].apply(clean_text)
```

In [ ]:

```
stop_words = set(stopwords.words('english'))  
  
def tokenize_text(text):  
    tokens = word_tokenize(text)  
    tokens = [t for t in tokens if t.isalpha() and t not in stop_words]  
    return tokens  
  
df['tokens'] = df['clean_text'].apply(tokenize_text)
```

In [46]:

```
lemmatizer = WordNetLemmatizer()  
  
def lemmatize(tokens):  
    return [lemmatizer.lemmatize(t) for t in tokens]  
  
df['lemmatized_tokens'] = df['tokens'].apply(lemmatize)
```

In [47]:

```
df[['text', 'lemmatized_tokens']].head()
```

Out[47]:

|   | text                                              | lemmatized_tokens                                 |
|---|---------------------------------------------------|---------------------------------------------------|
| 0 | From: mathew <mathew@mantis.co.uk>\nSubject: A... | [mathew, mathewmantiscouk, subject, altatheism... |
| 1 | From: lipman@oasys.dt.navy.mil (Robert Lipman)... | [lipmanoasysdtnavymil, robert, lipman, subject... |
| 2 | Newsgroup: comp.os.ms-windows.misc\ndocument_i... | [newsgroup, composmswindowsmisc, documentid, y... |
| 3 | Newsgroup: comp.sys.ibm.pc.hardware\ndocument_... | [newsgroup, compsysibmpchardware, documentid, ... |
| 4 | Newsgroup: comp.sys.mac.hardware\ndocument_id:... | [newsgroup, compsysmachardware, documentid, xo... |